Bavesh Raam S

github.com/baveshraam — linkedin.com/in/bavesh-raam-s

Computer Science student specializing in AI with a focus on evaluating and stress-testing machine learning systems. Experienced in designing evaluation frameworks, benchmarking model performance, particularly keen in deploying end to end Deep Learning Systems, RAG systems, computer vision models, and real-world ML deployments.

Education

B.Tech in Computer Science and Engineering (Artificial Intelligence) 2023–2027
Amrita Vishwa Vidyapeetham, Coimbatore CGPA: 8.09/10

Higher Secondary Education
Kamala Niketan Montessori School 83%

Experience

DRDO, Aeronautical Development Establishment, Bangalore April 2026 – June 2026
Deployed a production-grade, fully self-hosted agentic AI system for large-scale repository restructuring, combining centralized GPU-backed LLM serving, distributed semantic code intelligence, MCP service orchestration, and offline infrastructure automation. Implemented scalable code understanding and restructuring pipelines leveraging Serena, Tree-sitter, Clang-based analysis, cpp-tools, vector retrieval systems, and air-gapped deployment workflows supporting multi-user enterprise development environments..

Firmware Intern — Grace EV, Bangalore Nov 2025 – Dec 2025
Developed physics-based optimization models for EV motion control. Designed algorithms under real-world constraints (efficiency, latency, energy trade-offs) and validated behaviour across varying operating regimes.

Projects

Open Notebook — RAG System Evaluation Framework *LangChain, FAISS, FastAPI*
Designed evaluation framework for retrieval-augmented generation systems focusing on **retrieval fidelity, grounding, and reasoning consistency**.
Defined metrics including Hit@K, semantic relevance, and grounding alignment. Improved Hit@5 from **0.53** → **0.87** and answer quality from **3.1/5** → **4.4/5**.
Identified failure modes including hallucination under low-recall retrieval, sensitivity to chunking strategy, and breakdown in multi-hop reasoning. Conducted experiments isolating retrieval vs generation errors.

Small Vessel Detection — Model Behaviour and Tradeoff Analysis *PyTorch, Detectron2*
Evaluated detection models on SAR imagery dataset (**43,000 images**) using COCO metrics and scenario-based testing.
Compared architectures across accuracy–latency tradeoffs: ResNet-50-FPN (**0.73 mAP**) vs ResNet-18 (**0.68 mAP**, **2.3× faster**). Designed adaptive inference system selecting models based on scene complexity.
Analysed failure cases including missed detections in cluttered environments and sensitivity to noise patterns. Built hybrid system achieving **0.71 mAP** with real-time performance.

VisualVerse — NLP Pipeline Benchmarking and Evaluation *spaCy, scikit-learn*
Evaluated NLP pipelines for classification, keyphrase extraction, and topic modelling on **7,000 documents**.
Compared model families (Random Forest vs Naïve Bayes: **91% vs 74%**) and feature representations (Gradient Boosting vs TF-IDF: Precision@5 **0.68 vs 0.42**).
Analysed interpretability and output coherence through user studies (**4.3/5**) and identified limitations in semantic consistency across narrative structures.

Namma Vayal — AI-Powered Agricultural Platform *Flutter, TFLite, RAG*
Evaluated on-device crop disease detection model (EfficientNet-Lite0) under real-world deployment constraints, achieving **84% accuracy**, **120ms inference**, and **6MB model size** on mid-range devices.
Benchmarked retrieval pipelines for agricultural advisory queries using real-world datasets, improving relevance from **58%** → **87%** via hybrid search.
Analysed system behaviour under input variability (image noise, lighting conditions, and query ambiguity), identifying limitations in model generalisation and retrieval sensitivity.

Achievements

Finalist, Gen AI CBE Hackathon 2025 — Finalist, Infosys Global Hackathon 2025

Technical Skills

Machine Learning & Deep Learning: PyTorch, TensorFlow, scikit-learn, CNNs (ResNet, EfficientNet), Model Training, Feature Engineering, Model Evaluation
AI Systems & NLP: RAG Systems, LangChain, FAISS, NLP Pipelines, Information Retrieval
Data & Processing: NumPy, Pandas, Data Preprocessing, ETL Pipelines
Backend & Systems: FastAPI, Flask, Docker
Languages: Python, SQL, C/C++